

Raffay Arshad

Full Stack Engineer

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PROFESSIONAL SUMMARY

Full stack engineer experienced in building real-time products that people actually use, across web, desktop, and mobile. I like owning features from the database all the way up to the screen. Most of my recent work has been on AI search tools and live collaboration apps built with React, Next.js, Node.js, and PostgreSQL, with a lot of time spent on audio/video calling, messaging, and making things fast under load. Comfortable picking up new stacks and shipping without a lot of hand-holding.

TECHNICAL SKILLS

Languages: JavaScript, TypeScript, Dart, Kotlin, C/C++

Frameworks & Libraries: React.js, Next.js, Node.js, Express.js, Electron, Flutter

Real-Time & Streaming: WebSockets, Server-Sent Events (SSE), MQTT, LiveKit, Redis

Databases: PostgreSQL, MySQL, SQLite, NoSQL

Cloud & Tools: AWS (S3), Docker, Firebase, TanStack Query, Git

PROFESSIONAL EXPERIENCE

Full Stack Engineer

QLU.ai

Oct 2024 – Present

Islamabad, Pakistan

- Built and maintained scalable real-time web and desktop products with React, Electron, Node.js, Express, PostgreSQL, and WebSockets, delivering low-latency messaging, live collaboration, and persistent shared state across platforms.
- Architected and launched Zoro.ai, a standalone AI meeting assistant that auto-joins sessions through calendar sync and direct link injection, captures recordings to AWS S3, and generates AI notes, summaries, and transcripts.
- Engineered an AI-powered talent search experience that converts natural-language queries into structured, multi-step searches, streaming multi-modal results (reasoning, text, tables, charts) over SSE through an AI-driven filter loop that automatically derives and refines search criteria.
- Optimized API performance and data processing, enabling efficient handling of **80,000+ data items in a single API call** through algorithmic improvements and database optimizations, including query tuning, batching, and caching of hot aggregations.
- Built a client-facing collaborative workspace spanning desktop and web, featuring an Excalidraw-style interactive whiteboard and an embedded shared browser instance, with real-time co-editing and persistent state backed by PostgreSQL and AWS S3.
- Developed frontend and backend services for the QLU Social mobile app using Flutter, Node.js, Express.js, PostgreSQL, Redis, and SQLite, integrating Firebase Authentication and push notifications (FCM/APNs).
- Engineered real-time communication for QLU Social using LiveKit for audio/video calls and MQTT over WebSockets for group and direct messaging.
- Built QLU World, a next-generation virtual office mobile app in Flutter with BLoC state management on a Node.js/Express and PostgreSQL backend, enabling chat, audio/video calls, screen-sharing, and interactive workspaces.

Fellowship Associate – Full Stack Development

CodeHuddle

Aug 2024 – Oct 2024

Islamabad, Pakistan

- Developed dynamic frontend components in React and integrated backend services with Node.js, building responsive interfaces using Tailwind CSS and Next.js.
- Contributed to the flagship product “Gymyg” by writing robust unit tests and raising test coverage to safeguard code reliability and product quality.

Full Stack Developer

Funsol Technologies

May 2023 – Jun 2024

Islamabad, Pakistan

- Designed and shipped interactive dashboard modules in React and Node.js that surfaced product statistics, VPN usage analytics, and key business metrics, giving stakeholders clearer visibility into platform performance.

- Re-architected core data-fetching flows and optimized backend query handling, reducing load times and enabling faster, more reliable feature rollouts.
- Refactored shared frontend components into a reusable structure, improving maintainability and accelerating delivery of new dashboard views.
- Collaborated with backend and design teams to translate business requirements into a responsive, scalable application interface.

PROJECTS

Talent Intelligence Platform | *React, Node.js, PostgreSQL, SSE, TanStack Query*

- Architected a conversational AI talent platform that converts natural-language queries into structured, multi-step searches, streaming multi-modal answers (reasoning, text, tables, and charts) over SSE through an AI-driven filter loop that derives and refines search criteria automatically.
- Built the core streaming engine on a 15+ event typed protocol with dual-stream parallel generation, independent cancellation controls, and a candidate-feedback loop, rendering data token-by-token by caching streaming deltas in TanStack Query.
- Engineered the backend with infinite-query pagination against a custom scoring service, seamless cache-key migration to eliminate UI flicker during live streams, and normalized, transactional PostgreSQL persistence for full conversation histories.

QLU World | *Electron, React, Flutter, Node.js, Express, PostgreSQL*

- Engineered an AI-powered virtual office platform across web, desktop (Electron + React), and mobile (Flutter), featuring real-time messaging, audio/video calls, screen sharing, and collaborative workspaces on a scalable Node.js and PostgreSQL backend.
- Implemented LiveKit for ultra-low-latency communication, supporting instant “knock-to-join” connections, AI-assisted meeting scheduling, and immersive collaborative environments for remote teams.

Zoro.ai | *Next.js, Node.js, Express, PostgreSQL*

- Developed an automated bot orchestration system that synchronizes with user calendars and processes direct meeting links to autonomously join, record, and stream live sessions.
- Built a media pipeline using AWS S3 for AI processing handovers, integrated with a PostgreSQL-backed dashboard for transcript visualization and real-time “Edit with AI” refinement.

CERTIFICATIONS

AWS Academy Graduate – AWS Academy Cloud Foundations	2024
Web Development Masterclass – Online Certification Course	2024

PUBLICATIONS

Generative AI-Based Approach for Scalable and Cost-Effective Sustainable Urban Planning Using Satellite Imagery 2025